# GABRIELE BOZZOLA, PHD *Software Engineer at CliMA, California Institute of Technology*

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## IN A NUTSHELL

- Experienced **Python programmer** (6 years), author of the entire analysis and visualization library for Einstein Toolkit.
- Computational physicist and software developer with focus on **high-performance** and **scientific computing**.
- Innovative **researcher**, with 12+ publications, 100+ citations, $ 200k+ in grants, 2 awards for exceptional research.
- Specialized in **large-scale simulations**, with 10M+ CPU-hours awarded on national supercomputers.
- Passionate instructor and **communicator**, with 30+ invited/contributed talks, 1 award for excellence in teaching.
- Active member of the open-source world, co-maintainer of vterm–the fastest terminal emulator for Emacs.

## SKILLS

MY DAY-TO-DAY: Julia, Python (NumPy, matplotlib, SciPy, tests, documentation, packaging), Emacs, git, Bash, Linux, simulations, High-Performance Computing, CI, benchmarking, differential equations, differential geometry.

EXPERIENCE WITH: C, C++03, Fortran90, awk, regexp, OpenMP, High-Throughput Computing, HTCondor, PBS, MPI.

## EDUCATION

2018–2023 University of Arizona — MSc and PhD in Astrophysics (GPA: 4/4, PhD defense: April 2023)

2012–2017 University of Milan — BSc and MSc in Physics (*summa cum laude*, GPA: 30/30)

## RELEVANT EXPERIENCE

- Reduced time to implement new analyses and train new Einstein Toolkit users by $> 90\%$ by designing and developing a new Python library for **post-processing and visualization** (kuibit, >30k lines of code).
- Advanced theoretical models of black holes by **designing and performing massively-parallel simulations** (∼1000 cores) with novel numerical methods for differential equations and data analysis. Disseminated results through 12+ peer-reviewed publications (one selected as journal cover) and 30+ contributed/invited talks.
- Enabled new scientific capabilities and found new results by building modules for initial data, data processing, diagnostics, and infrastructure for the Einstein Toolkit, the most popular public code for relativistic astrophysics. Worked on a **large legacy code base** (> 500k lines of C/Fortran code) and in a **distributed and multidisciplinary community**. Took part in planning, documenting, testing, and releasing new versions.
- Performed scaling tests comparing infiniband with RoCE and determined impact on large MPI simulations as part of the **benchmarking** team that advised UArizona on the purchase of its latest ($ 2M) supercomputer.
- Discovered new neutron star properties by efficiently analyzing ∼100k models with high-throughput computing.
- Tested and **optimized** Einstein Toolkit on a non-x86 architecture, leading to improved support for IBM POWER9.
- Expanded capabilities of a GPU code for general-relativistic ray tracing by implementing new physics modules.
- Characterized properties of a new design for radiation-hardened integrated circuits by running simulations with Cadence Virtuoso&Spectre (for my BSc thesis, within the CHIPIX65 collaboration).
- Developed and maintained **open-source packages** for science and Emacs.

## WORK EXPERIENCE

- **Software Engineer** at the California Institute of Technology (CA). Aug 2023 –
- **Research and Teaching Assistant** at University of Arizona in Tucson (AZ). Jan 2018 – Aug 2023
  - Taught classes, mentored 5 students, and received 3 awards for excellence in teaching and research.
  - Reviewed publications for international academic journals (including the Journal of Open Source Software).
  - Co-founded and ran for two years the Steward Observatory International Scholars Task Force that assisted >20 international scholars in their transition to UArizona by providing mentorship and additional resources.

## RELEVANT GRANTS AND AWARDS

2021 Selected as one of 21 NASA Future Investigator in Space Science ($ 135k + 146k node-hours valued $ 69k)

2020 Selected as one of 5 Texas Advanced Center for Computing Frontera Fellows ($ 44k + 50k node-hours)

**LANGUAGES AND NATIONALITY:** Italian (mother tongue), English (full proficiency); Italian nationality.